

CML — BCR::ABL1 Biology, Transcripts & Progression



1. Philadelphia chromosome

WHAT YOU SEE

PHILADELPHIA banner over an industrial reactor.

WHAT IT ENCODES

The Philadelphia chromosome is the derivative chr 22 from t(9;22)(q34;q11), present in >90–95% of CML. It fuses BCR (chr 22) to ABL1 (chr 9), creating the constitutively active BCR::ABL1 tyrosine kinase.

BOARD TRAP

Ph chromosome also appears in some ALL and rare AML — in CML it is near-universal; its absence should make you reconsider the diagnosis (or use FISH/PCR for cryptic fusions).

2. BCR::ABL1 fusion gene

WHAT YOU SEE

Chromosomes 9 and 22 swapping segments, BCR::ABL1, >90%.

WHAT IT ENCODES

Reciprocal translocation moves ABL1 onto BCR. The fusion gene encodes a kinase locked in the active state, driving uncontrolled myeloid proliferation independent of growth signals.

BOARD TRAP

It is a RECIPROCAL translocation — both der(9) and der(22) form; the der(22) Ph chromosome carries the oncogenic fusion.

3. Constitutive KINASE

WHAT YOU SEE

Glowing KINASE core; ATP/myristoyl lock LOST.

WHAT IT ENCODES

BCR::ABL1 lost its autoinhibitory myristoyl regulation, so the ABL1 kinase is constitutively on. TKIs (imatinib etc.) block the ATP-binding pocket; asciminib uniquely binds the myristoyl pocket (STAMP inhibitor).

BOARD TRAP

Loss of myristoyl autoinhibition explains why asciminib — an allosteric myristoyl-pocket binder — works even against ATP-pocket mutations like T315I.

4. Downstream RAS/MAPK/STAT/PI3K/MYC

WHAT YOU SEE

Pipes labeled RAS, MAPK, STAT, PI3K, MYC firing off the kinase.

WHAT IT ENCODES

BCR::ABL1 activates RAS-MAPK, JAK-STAT (STAT5), PI3K-AKT and upregulates MYC — driving proliferation, blocking apoptosis, and reducing stromal adhesion.

BOARD TRAP

These survival pathways are why deep responses still leave quiescent leukemic stem cells — they survive off-target of kinase inhibition, the rationale for not stopping TKIs prematurely.

5. p210 (e13a2/e14a2)

WHAT YOU SEE

Large p210 drum, e13a2/e14a2, PCR PROBE.

WHAT IT ENCODES

The major breakpoint (M-bcr) yields p210 (b2a2/e13a2 or b3a2/e14a2), the transcript in >95% of classic CML. This is the transcript monitored quantitatively on the International Scale.

BOARD TRAP

Standardized IS PCR monitoring assumes p210; atypical transcripts (e.g., e1a2) need a different assay and can be 'falsely negative' on standard probes.

6. p190 (e1a2) — Ph+ ALL

WHAT YOU SEE

Smaller p190 drum, ALL, Ph+ worse in CML.

WHAT IT ENCODES

Minor breakpoint (m-bcr) yields p190 (e1a2), typical of Ph+ ALL and a minority of CML. p190 CML behaves more aggressively (more monocytosis, worse prognosis).

BOARD TRAP

p190 is the classic Ph+ ALL transcript; finding p190 in 'CML' flags a more aggressive monocytic phenotype, not a benign variant.

7. Indolent p230 (e19a3)

WHAT YOU SEE

Small INDOLENT vial, e13a3/e14a3 / e19a3.

WHAT IT ENCODES

The micro breakpoint (μ-bcr) yields p230 (e19a2/e19a3), associated with the rare neutrophilic-CML variant that runs a more indolent course with prominent neutrophilia and sometimes thrombocytosis.

BOARD TRAP

p230 neutrophilic CML can be mistaken for chronic neutrophilic leukemia — but CNL is BCR::ABL1 NEGATIVE and CSF3R-mutated.

8. Falsely negative transcript

WHAT YOU SEE

Red X, FALSELY NEGATIVE.

WHAT IT ENCODES

If the patient's transcript is atypical (e.g., e1a2/p190 or e19a3/p230), a PCR assay designed only for p210 can read falsely negative. The breakpoint must be characterized at diagnosis.

BOARD TRAP

A 'negative' BCR::ABL1 PCR in a clinically obvious CML usually means a non-p210 transcript, not absence of disease — characterize the breakpoint up front.

9. Second hit / progression

WHAT YOU SEE

SECOND HIT lightning, 25% NORMAL.

WHAT IT ENCODES

Chronic-phase CML is BCR::ABL1-driven; progression to accelerated/blast phase requires secondary genetic hits (additional clonal abnormalities, TP53/RUNX1 mutations). ~25% of marrow may remain normal hematopoiesis early.

BOARD TRAP

BCR::ABL1 alone causes chronic phase; blast crisis needs a 'second hit' — this is why ACAs predict transformation.

10. ACAs of progression

WHAT YOU SEE

Bottom: DOUBLE Ph, +8, iso17, 20q-, i(17q), 3q26.2 → TRANSFORMATION.

WHAT IT ENCODES

High-risk additional cytogenetic abnormalities driving transformation include double Ph, trisomy 8, isochromosome 17q (i(17q)/TP53 loss), 3q26.2 (EVI1/MECOM) and complex karyotypes. They define accelerated phase and predict blast crisis.

BOARD TRAP

i(17q) means functional TP53 loss and a poor prognosis — the appearance of any major-route ACA upgrades the patient toward accelerated/blast phase even with controlled counts.

11. Lineage potential

WHAT YOU SEE

Bottom-left: myeloid, erythroid, megakaryo, monocyte; B-cell, T-cell, stromal struck through.

WHAT IT ENCODES

CML arises in a pluripotent hematopoietic stem cell, so the Ph clone can populate myeloid, erythroid, megakaryocytic and monocytic lineages (and B/T in blast crisis). Stromal cells are NOT part of the clone.

BOARD TRAP

Stromal/marrow fibroblasts are reactive, not clonal — myelofibrosis in CML is a secondary reactive process, not BCR::ABL1-bearing stroma.
